

DAY-1 SYSTEM REQUIREMENTS & PRODUCT SPECIFICATION

UNCHARTED ATLAS

India's Free Investment Intelligence Terminal

Prepared for: Chief Technology Officer

Document type: Product & System Requirements Sheet (v1.0)

Scope: Founding build — no budget constraint, top-tier talent, best-available technology

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1. Vision & Mandate

Uncharted Atlas is being built as India's first fully free, retail-grade investment intelligence terminal — a single destination that brings together real-time market data, institutional-grade portfolio analytics (drawn from the CFA Level I body of knowledge), multi-broker portfolio aggregation, and news/sentiment intelligence, with zero paywall for core analytics. The founding mandate for this build:

- **No cost ceiling on Day 1.** The CTO is authorized to select best-in-class infrastructure, data vendors, and talent. Cost optimization and break-even modelling are explicitly out of scope for this document — that is a Phase 2 exercise.
- **One brand, one name.** Every surface — product, domain, app store listing, social handles, press mentions — carries a single identity: **Uncharted Atlas**. No sub-brands, no "powered by" co-branding on the primary UI.
- **The USP is analytical depth, not just data.** Real-time prices are commoditized in India (five apps already do this free). The differentiator is packaging CFA-grade portfolio analytics — Sharpe/Sortino/Treynor/M², factor exposure, duration/convexity, Greeks, DuPont decomposition — into a live, visual, zero-cost product. This is common in the US (Bloomberg Terminal, Portfolio Visualizer, YCharts, Seeking Alpha Premium) but effectively absent, free, and unified in India.

Note on scope of this document: You mentioned wanting to integrate an API referred to as "Traction" for company/financials data and mentioned demat/broker login integration. I could not verify "Traction" as a named financial data vendor — it may be a mishearing/autocorrect of a vendor name (e.g. Refinitiv, TickerTape, TrendlyneAPI, Truedata, or a Bloomberg-adjacent feed). Section 4 below lists the realistic vendor shortlist across both market-data and broker-API categories so the CTO can confirm the exact vendor with you before contracting.

2. What Exists Today vs. What Uncharted Atlas Must Deliver

This table frames the competitive gap: features common in US-facing retail platforms that are either absent, fragmented, or paywalled in India.

Capability	Typical US Access	Indian Market Today	Uncharted Atlas (Free)
Multi-metric risk-adjusted return scoring (Sharpe, Sortino, Treynor, M ² , Jensen's alpha, Information Ratio)	Portfolio Visualizer, Morningstar Premium (\$\$\$)	Not offered free on any major retail app	Live, per-holding and per-portfolio, recalculated on every price tick
Options Greeks & payoff visualizer (delta, gamma, theta, vega, rho)	Thinkorswim, Tastytrade — free with brokerage	Sensibull (freemium), NSE Option Chain (raw data only)	Full Greeks + payoff diagrams + IV surface, free, broker-agnostic
Bond duration/convexity & yield-curve analytics	Bloomberg Terminal, YCharts (\$\$\$\$)	Virtually absent for retail	Free duration/convexity calculator + live G-Sec & corporate yield curve
Six-pane, Bloomberg-style customizable dashboard	Bloomberg Terminal (\$2,000+/mo)	Not available at any price point for retail	Free, six-tile drag-and-drop workspace (Section 6)
Multi-broker unified portfolio (single login, all demat accounts)	Fragmented even in the US	Not available — each broker is siloed	OAuth-based broker aggregation vault (Section 7)
Factor/smart-beta screener with live DuPont & activity-ratio decomposition	Zacks, Portfolio123 (\$)	Screener.in (fundamentals only, no risk analytics)	Free, integrated with portfolio-level analytics
News + sentiment fused directly onto holdings & correlation matrix	Bloomberg News, Benzinga Pro (\$\$)	News and portfolio views are separate apps	Unified news-to-holding sentiment tagging

3. Platform Architecture — Day 1 Requirements

3.1 High-Level Architecture Principles

- **Event-driven, microservices-first.** Independent services for: market-data ingestion, analytics/computation engine, portfolio ledger, broker-integration gateway, news/NLP engine, user/auth, notification, and the frontend BFF (backend-for-frontend).
- **Real-time by default.** All price, portfolio-value, and analytics tiles must update via streaming (WebSocket/SSE), not polling.
- **Separation of "market data" from "user financial data."** User holdings, broker credentials/tokens, and PII sit in a hardened, separately audited data boundary from public market-data caches.
- **Computation-heavy analytics (Greeks, duration/convexity, factor regressions, Monte Carlo) run server-side** in a dedicated quant engine, not client-side, to guarantee consistency and enable caching/replay.

3.2 Recommended Technology Stack (best-of-breed, cost unconstrained)

Layer	Recommendation
Frontend (web)	Next.js (React 18+, App Router) + TypeScript; TanStack Query for data fetching/caching; Zustand/Redux Toolkit for state; WebSocket client via native browser API or Socket.IO; charting via TradingView Lightweight/Charting Library (licensed) + D3.js for bespoke analytics visuals (correlation heatmaps, factor exposure radar, Greeks surfaces).
Frontend (mobile)	React Native (or native Swift/Kotlin if budget truly unconstrained and a "no-compromise" native feel is desired) sharing the design system and analytics SDK with web.
API layer	GraphQL gateway (Apollo/Hasura) for flexible client queries + dedicated low-latency gRPC/WebSocket channel for streaming price/analytics ticks.
Core backend services	Go or Rust for latency-critical services (market-data ingestion, streaming fan-out, Greeks/duration engine); Python (FastAPI) for the quant/analytics and ML/NLP services where the CFA-curriculum calculation libraries and data-science ecosystem (NumPy, SciPy, statsmodels, QuantLib) live; Node.js/NestJS acceptable for user/account/broker-integration services.
Streaming backbone	Apache Kafka (or Redpanda) for market-data and event streaming; Kafka Streams/Flink for real-time analytics recomputation (rolling Sharpe/beta/correlation as new ticks arrive).
Time-series storage	ClickHouse or TimescaleDB for tick/OHLC history; QuestDB as an alternative optimized for financial tick data.
Operational database	PostgreSQL (multi-AZ) for user accounts, portfolios, broker-token metadata, watchlists.
Cache / real-time state	Redis (or KeyDB) for session state, computed-metric cache, and pub/sub fan-out to WebSocket gateways.
Search	Elasticsearch/OpenSearch for instrument search, news search, and full-text filing search.
ML / NLP	Python stack (PyTorch/HuggingFace transformers) for news sentiment tagging, earnings-call summarization, and the "financial red-flag" anomaly detector (Section 5.5); vector database (Pinecone/Weaviate/pgvector) for semantic news-to-holding matching.
Cloud & infra	AWS or GCP, multi-AZ within Mumbai/Hyderabad region for latency to NSE/BSE, with a secondary region for DR. Kubernetes (EKS/GKE) for all services; Terraform for infra-as-code; Istio/Linkerd service mesh for observability and mTLS between services.
Observability	Grafana + Prometheus + Loki/Tempo (or Datadog if a managed, best-in-class single pane is preferred given no budget constraint).
CI/CD & security scanning	GitHub Actions/GitLab CI, with SAST/DAST (Snyk, Semgrep), secrets scanning, and container image scanning (Trivy) on every merge.

4. Data & API Integration Requirements

4.1 Market Data & Fundamentals Vendors (shortlist to confirm)

Vendor category	Candidates	What they provide
Exchange-licensed real-time feed (India)	NSE Data & Analytics (NSE DnA), BSE Market Data, or a licensed re-distributor such as Truedata, GlobalDataFeeds, or Kite Connect market feed	Live tick/quote data, market depth, indices — required for any real-time price display in India under exchange data-redistribution rules.
Fundamentals / financial statements	Tickertape API, Trendlyne, screener.in (no public API — would need a data licensing conversation), Refinitiv Eikon/Datastream, S&P Capital IQ, Morningstar Direct	Company financials, ratios, ownership/shareholding data, corporate actions.
Global equities/macro (for the "US-style" benchmark views)	Refinitiv, Bloomberg B-PIPE/Enterprise (if budget-unconstrained and enterprise licensing is viable), Polygon.io, IEX Cloud, Alpha Vantage	Cross-market comparison, global index correlation, FX and commodity feeds for the Economics module.
News	PTI, Reuters News API, Benzinga, or an aggregated RSS + licensing deal with Indian financial media (Moneycontrol, Livemint, Business Standard)	Real-time news for the sentiment-tagging engine.

Confirm with the CTO/legal team the exact commercial data-redistribution license required — NSE/BSE both require a data-vendor agreement before any real-time price can be shown to end users, even for free products. This is a Day-1 legal dependency, not just a technical one.

4.2 Broker / Demat Aggregation — Correct Technical Approach

You described wanting users to "log into their demat account" and have "the API key entered automatically" so the platform can manage their portfolio without switching apps. The technically and legally sound way to build this (and the way every credible aggregator — Smallcase, Groww's connect features, or US equivalents like Plaid for brokerage — does it):

- **Never store a user's raw broker username/password.** Each broker (Zerodha Kite Connect, Upstox API, Angel One SmartAPI, ICICI Direct Breeze, Fyers API, 5paisa, Dhan API) exposes an official OAuth-based Connect API. The user authenticates directly on the broker's own login screen; Uncharted Atlas only receives a scoped, revocable access token.
- **Token vault:** a dedicated, encrypted (HSM-backed, e.g. AWS KMS/CloudHSM) service stores only these tokens, never credentials, with automatic rotation/expiry handling per broker's API terms.
- **"Credits for API access" model:** rather than the platform holding a shared master API key, each user connects their own broker account (most Indian broker APIs are free or nominal for personal use, e.g. Kite Connect's personal API token). Uncharted Atlas's "membership credits" can instead fund *premium data/analytics compute* (e.g., deeper history, faster refresh, extra watchlists) — not the brokerage connection itself, which should remain a direct, user-authorized OAuth link for compliance reasons.
- **SEBI Account Aggregator (AA) framework:** India already has a regulated framework (Sahamati/AA ecosystem, RBI-licensed AAs like Perfios, Finvu, OneMoney) purpose-built for consent-based aggregation of financial account data. Integrating with an AA is the compliant path to "one login, all accounts" rather than building bespoke credential storage.

USP angle: "One dashboard, every broker" — a user with accounts across Zerodha, Groww, and Upstox sees one consolidated portfolio, one correlation matrix, one risk score — something no single Indian broker app currently offers, built entirely on top of consent-based OAuth/AA rails (not credential storage), which also removes a major security liability for the platform.

5. Core Analytics Modules (mapped to CFA Level I curriculum)

These modules translate CFA Level I learning content directly into live, visual, always-on product features — this is the primary USP layer of Uncharted Atlas.

5.1 Portfolio & Risk Analytics Module (from Quantitative Methods / Portfolio Management)

Metric / Feature	Product Requirement
Sharpe Ratio	Live, per-holding and whole-portfolio, benchmarked against Nifty 50/Sensex and a user-selected peer index.
Sortino Ratio & Roy's Safety-First Ratio	Downside-deviation-based scoring, useful for conservative retail investors — flagged as a differentiator versus Sharpe-only apps.
Treynor Ratio & Jensen's Alpha	Beta-adjusted return attribution versus benchmark; requires a rolling regression engine (see 5.6).
M-squared (M ²) Measure	Risk-adjusted return rescaled to benchmark volatility, displayed alongside Sharpe so retail users can compare portfolios of different risk levels on a like-for-like basis — a metric essentially unseen in Indian retail apps.
Information Ratio & Tracking Error	For users benchmarking against an index or smallcase-style model portfolio.
Portfolio Turnover Ratio	Auto-calculated from the user's transaction history (post broker-sync) — flags high-churn behavior with an educational nudge.
Correlation & Covariance Matrix	Live heat-map across all portfolio holdings and against major indices/asset classes — one of the requested "showcase" features (Section 6).
Monte Carlo Portfolio Simulation	Forward-looking distribution of outcomes for retirement/goal planning, using historical return/volatility inputs.
Active Share & Factor Exposure	Decomposes a portfolio's value/growth/quality/momentum/size tilts versus benchmark.

5.2 Equity Module (from Equity Investments)

- Valuation suite: P/E, P/B, EV/EBITDA, Dividend Discount Model and Free-Cash-Flow-to-Equity calculators with editable growth assumptions.
- Market-efficiency indicators and screener flags (e.g., momentum/reversal signals framed as educational, not investment advice).
- Industry & peer-comparison view, auto-generated from fundamentals data.

5.3 Fixed Income Module (from Fixed Income)

- Live G-Sec and corporate bond yield curve visualizer (spot, par, forward curves).
- Modified duration, effective duration, and convexity calculators per bond and at the portfolio level.
- Credit-spread analysis versus benchmark yield curve; callable/puttable bond adjustment tools.

5.4 Derivatives Module (from Derivatives)

- Live options chain with full Greeks (delta, gamma, theta, vega, rho) recalculated on every tick.
- Interactive payoff-diagram builder for multi-leg strategies.
- Put-call parity checker (flags arbitrage-style mispricings for educational purposes) and implied-volatility surface view.

5.5 Financial Statement Analysis Module

- Full ratio suite: liquidity (current, quick), solvency (debt-to-equity, interest coverage), profitability (ROE, ROA, margins), and activity/turnover ratios (inventory turnover, receivables turnover, total asset turnover).
- DuPont ROE decomposition (3-step and 5-step) with historical trend charts.
- Common-size statement viewer and an ML-based "financial red-flag" detector (aggressive revenue recognition, deteriorating cash conversion, rising related-party transactions) — flagged in Section 8 as a strong USP.

5.6 Economics Module

- Macro dashboard: GDP growth, inflation (CPI/WPI), repo rate, business-cycle phase indicator, INR exchange-rate module.
- Economic calendar with market-impact tagging (RBI policy, Union Budget, US Fed decisions).

5.7 Trust & Conduct Layer (inspired by the Ethics component of the curriculum)

Rather than a user-facing "ethics module," this becomes the platform's own governance backbone — directly addressed to the CTO as a build requirement:

- No paid stock tips or sponsored "buy" signals anywhere in the analytics UI; any sponsored content must be clearly separated and labeled.
- Full model/algorithm transparency pages explaining how each score (Sharpe, M², red-flag detector, etc.) is calculated, so users can audit the numbers.
- Clear, persistent disclaimers that Uncharted Atlas provides analytics/education, not personalized investment advice — a SEBI Research Analyst/Investment Adviser registration question to resolve with legal counsel before any "buy/sell" style recommendation is ever shown.

6. Real-Time Dashboard & UX Requirements ("The Six-Window View")

- **Six-pane customizable workspace** as the flagship home screen — modelled on the multi-monitor feel of a professional terminal, delivered in a single browser/app window. Each of the six panes is independently configurable from a widget library: live watchlist, price chart, correlation heat-map, portfolio risk-metric scorecard (Sharpe/Sortino/M²), options Greeks panel, news+sentiment feed, sector heat-map, economic calendar, DuPont/ratio breakdown.
- Drag-to-resize and drag-to-rearrange panes; saved layouts per user (and shareable layout templates, e.g. "Options Trader," "Long-Term FI Investor," "Index Tracker").
- Dark, terminal-inspired theme by default (navy/charcoal with high-contrast data-viz colors) with a light theme option.
- Global command palette (type a ticker or metric name, jump straight to it) — a well-loved Bloomberg-terminal-style interaction.
- Unified news feed with per-headline sentiment tags (positive/neutral/negative) auto-linked to any portfolio holding it mentions.

USP angle: No free Indian platform currently offers a persistent, multi-pane, drag-and-drop "terminal" workspace — most are single-view mobile-first apps. This is the single most visible differentiator worth building first.

7. Membership, Credits & Broker-Login Section

- Free core tier: full access to all analytics modules in Section 5, six-pane dashboard, and single-broker OAuth connection.
- Credit-based premium tier (funds infrastructure, not the brokerage link itself): extended historical data, faster refresh intervals, multi-broker aggregation across more than one account, priority news/NLP processing.
- Unified broker-connect screen listing all supported brokers (Section 4.2), each routing to that broker's own OAuth login — never a Uncharted Atlas-hosted credential form.
- A visible, revocable "Connected Accounts" settings page so users can see and disconnect any broker link at any time — this is both a trust feature and generally an expectation under India's Digital Personal Data Protection (DPDP) Act 2023.

8. Ranked USP Summary — What to Build First

Rank	Feature	Why it's a genuine gap in India
1	Six-pane live terminal dashboard	No free Indian retail platform offers a persistent, customizable multi-pane workspace.
2	Full risk-adjusted return suite (Sharpe, Sortino, Treynor, M ² , Jensen's alpha) live on every portfolio	These sit behind paid tools (Morningstar, Portfolio Visualizer) even in the US; in India they are essentially unavailable to retail.
3	Multi-broker unified portfolio via OAuth/Account-Aggregator	Every Indian broker app is siloed; no consolidated, compliant cross-broker view exists today.
4	Live options Greeks + payoff builder, free	Partially addressed by Sensibull (freemium); a fully free, integrated version is a gap.
5	Bond duration/convexity & yield-curve tools	Effectively nonexistent for retail investors in India today.
6	ML-based financial-statement red-flag detector	No mainstream Indian retail app auto-flags earnings-quality or balance-sheet risk.

9. Non-Functional Requirements

Category	Requirement
Latency	Sub-second price-tick to UI update; sub-2-second recomputation of portfolio-level analytics on any holding change.
Availability	99.9%+ uptime target for market hours (9:00 AM–4:00 PM IST) with automated failover across AZs.
Security	End-to-end encryption in transit and at rest; HSM-backed token vault; SOC 2-equivalent controls; regular third-party penetration testing before public launch.
Compliance	DPDP Act 2023 data-privacy compliance; NSE/BSE data-redistribution licensing; SEBI Research Analyst/Investment Adviser registration review before any "recommendation"-style feature ships; RBI Account Aggregator framework alignment for broker connectivity.
Scalability	Architecture must handle national-scale concurrent usage spikes (market open/close, budget day, results season) without degradation.
Auditability	Every analytics calculation must be traceable/reproducible (versioned calculation engine) to support the "algorithm transparency" trust feature in Section 5.7.

10. Team Composition (Day 1)

- 1 CTO / Head of Engineering (overall architecture ownership)
- 2–3 backend engineers (Go/Rust) — market-data ingestion & streaming
- 2 quant/data engineers (Python) — analytics engine (Sections 5.1–5.6)
- 2 frontend engineers (React/Next.js) — dashboard & widget system
- 1 ML/NLP engineer — news sentiment & red-flag detector
- 1 DevOps/SRE — infra, observability, security posture
- 1 security engineer — token vault, compliance, penetration-test coordination
- 1 product designer — terminal-grade UI/UX and design system
- 1 compliance/legal lead — SEBI/RBI/DPDP coordination (critical path before public launch, independent of engineering timeline)

11. Branding & SEO Requirements — "Uncharted Atlas"

- Single-brand rule: domain, app names, and all social handles should match exactly (e.g., unchartedatlas.com/.in), with no sub-brand names appearing above the fold anywhere in-product.
- Consistent meta title pattern across pages, e.g. "Uncharted Atlas — Free Portfolio Analytics & Real-Time Market Terminal for India," with unique, descriptive meta descriptions per module page (Equity, Fixed Income, Derivatives, etc.) so each ranks independently for its own search terms.
- Dedicated, content-rich landing pages per analytics concept (e.g., "What is the Sharpe Ratio," "M-squared Explained") to capture long-tail educational search traffic — this doubles as user education and organic SEO.
- Structured data (schema.org FinancialProduct/Article markup) on educational and market pages to improve rich-result eligibility.
- Fast Core Web Vitals (sub-2.5s LCP) — critical both for SEO ranking and for a "terminal-grade" feel.

12. Suggested Build Sequence

1. **Phase 0 (Weeks 1–4):** Architecture setup, data-vendor contracts (Section 4), broker OAuth integrations, core auth/user system.
2. **Phase 1 (Weeks 5–12):** Six-pane dashboard shell, live pricing/watchlists, Portfolio & Risk Analytics module (5.1), single-broker sync.
3. **Phase 2 (Weeks 13–20):** Fixed Income, Derivatives, and FSA modules; multi-broker aggregation; news/sentiment engine.
4. **Phase 3 (Weeks 21–28):** Economics module, ML red-flag detector, factor/smart-beta screener, SEO content build-out, security audit & penetration test, compliance sign-off, public launch.